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When does resource recovery beat a sanitary landfill? A provenance-aware screening LCA and TEA of six municipal solid waste pathways for 21 Indonesian cities and regencies

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Abstract

Indonesian local governments must choose between sanitary landfilling and recovery pathways with waste data that are sparse, of different years and of uneven definition. We present an open, provenance-aware screening model that compares six options per tonne of mixed municipal solid waste (MSW) at the facility gate in 2025: sanitary landfill with flare, waste-to-energy (WtE), refuse-derived fuel (RDF) for cement kilns, anaerobic digestion (AD) of food waste, polyhydroxybutyrate (PHB) from landfill gas, and integrated RDF + AD. All pathways share one LCA and TEA boundary, including residue disposal. Composition data from local master plans (RIPS) of 21 cities and regencies were harmonised to a 2025 baseline with a single, audited projection of tonnage, and every value was labelled as local measurement, literature default, proxy, assumption, scenario or gap. Feasibility gates, Pareto screening and the probability of being best over a sampled carbon value (0–100 USD/t CO₂e) were combined with Monte Carlo simulation. At central values sanitary landfill with flare had the lowest social cost in all locations up to 25 USD/t CO₂e, whereas RDF + AD was best in 18 locations at 50 USD/t and 20 at 100 USD/t, with an abatement cost of 42 (25–62) USD/t CO₂e against landfill. WtE was preferred only with the Perpres 109/2025 tariff in three large locations, AD only for source-separated food waste, and PHB nowhere. Scenario discovery over 40,000 combinations of conditions condensed these results into decision rules: for mixed waste, landfill with flare is preferred below about 40 USD/t CO₂e and wherever no cement kiln lies within about 300 km, RDF + AD above about 55 USD/t with a kiln in reach, WtE only with the tariff, at least 1,000 t/day and an LHV of at least 7 MJ/kg, and AD where food is separated at source. RDF + AD and WtE also saved the most fossil energy (about 4 GJ/t), and every recovery option reduced land take relative to landfilling. The ranking between landfill and RDF + AD was robust to the tonnage-year and data-gap scenarios but not to the moisture assumption: with IPCC default moisture instead of as-received values, RDF + AD became most probable in 20 of 21 locations. Measured as-received moisture and heating value, and a common definition of managed waste, are therefore the data that would most improve such decisions. The model is released as a Google Colab tool that researchers and planners can run for their own cities.

Highlights

- Six MSW options compared per tonne within one LCA/TEA boundary for 21 Indonesian locations
- Every input labelled as measurement, default, proxy, assumption, scenario or gap
- Landfill with flare is cheapest below 25 USD/t CO₂e; RDF + AD leads at 50–100 USD/t
- WtE needs the Perpres 109/2025 tariff and 1,000 t/day; PHB is not preferred anywhere
- Decision rules from scenario discovery state when each option wins
- The moisture assumption, not tonnage year or data gaps, can reverse the ranking

Keywords: municipal solid waste; life cycle assessment; techno-economic assessment; refuse-derived fuel; anaerobic digestion; data provenance; scenario discovery; Indonesia

1. Introduction

Comparing waste-treatment options per tonne of waste, with credits for the energy and materials they displace, is established practice in waste LCA (Laurent et al., 2014; Astrup et al., 2015; Khandelwal et al., 2019). Such comparisons are informative for decisions only if three conditions hold: the options deliver the same function

within equivalent system boundaries (ISO, 2006), every assumption is stated, and uncertainty is carried to the conclusion (Clavreul et al., 2012; Bisinella et al., 2017). The first condition is often violated when recovery pathways treat only part of the waste and the remainder silently leaves the system; the third is weakened when composition, which is itself uncertain, is treated as known (Edjabou et al., 2017; Bisinella et al., 2017).

For Indonesian cities and regencies a fourth condition becomes critical: the provenance of the data. Local waste master plans (Rencana Induk Pengelolaan Sampah, RIPS) report composition and tonnage measured in different years with different category definitions, tonnage totals whose reference year is not always stated, and managed-waste indicators that use different numerators and denominators. National policy meanwhile creates scale-dependent incentives: Perpres 109/2025 grants a WtE electricity tariff to facilities of at least 1,000 t/day. Fraction-level properties such as moisture, heating value and degradable organic carbon (DOC) are rarely measured locally, so models fall back on global IPCC defaults (IPCC, 2006, 2019) or on values from other countries, for example the detailed Danish household-waste characterisation of Edjabou et al. (2015). When such values enter a model without a status label, a proxy can be mistaken for a measurement and an analyst's assumption for a city observation.

This study asks under what Indonesian city and waste-system conditions each recovery pathway becomes preferable, and which data limitations could change the answer. Its contributions are: (i) a screening model in which six options manage the whole tonne within one boundary, with sanitary landfill as both a baseline and an option; (ii) a reproducible data layer that harmonises 21 RIPS records to a 2025 baseline with a single, audited tonnage projection and labels every value by its evidence type; (iii) a three-level waste-fraction taxonomy with explicit parameter gaps, kept separate from the computation so that unverified allocation priors cannot influence results; and (iv) a decision analysis that reports the probability of each option being best over a range of carbon values, together with data scenarios that isolate the effect of individual provenance decisions; and (v) scenario discovery that turns the model into explicit decision rules, released with the model as a reproducible Google Colab tool for other cities.

2. Materials and methods

2.1 Goal, scope and functional unit

The functional unit is the management of 1 t of mixed MSW, wet weight, as received at the gate of the treatment facility, with the composition of the location, in 2025. Six options are compared: sanitary landfill with gas collection and flare (SL), WtE by grate incineration (S1), RDF co-processed in the nearest cement kiln (S2), wet AD of food waste separated from mixed waste (S3), PHB produced from captured landfill gas (S4) and an integrated plant in which an RDF line also separates food for a digester (S5). Every option includes its process energy and materials, direct emissions (fossil CO₂, CH₄, N₂O), the landfill of its residues (ash, rejects, uncaptured waste, digestate), RDF haulage and the credit for the product it displaces (grid electricity, coal, polypropylene). Collection, capital goods, biogenic CO₂, landfill carbon storage and recycling credits are excluded for all options. Results are expressed against two baselines, an open dump (OD) and a sanitary landfill, as $\Delta G_k = G_0 - G_k$. Climate change is assessed with GWP100 of IPCC AR6 (non-fossil CH₄ 27.0, N₂O 273), GWP20 being a sensitivity case. Costs are net levelised costs in USD per tonne (AACE class 5).

2.2 Study locations and data harmonisation to 2025

The 21 locations comprise 5 cities (kota) and 16 regencies (kabupaten) in West Sumatra, West Java, Banten, Central Java, East Java, Bali and East Kalimantan, with 2025 tonnage from 50 to 2,891 t/day. For each location three years are stored separately: the composition sampling year t_c (2014–2025), the tonnage source year t and the baseline year. Tonnage is projected once,

$$Q_{2025} = Q_t (1 + g)^{2025 - t} \quad (1)$$

with the RIPS growth rate g where available and otherwise the median of the six reported rates (1.73% yr⁻¹). Domestic and non-domestic tonnage use the same factor. The model parser reads Q_{2025} directly and asserts that the tonnage it uses equals the database value, which prevents a second projection. Eight totals come from an overview index whose reference year could not be verified; they are used as reported, their year is left blank, and a projection from the composition year (up to +21%) is run as a scenario. Composition is used as a proxy for 2025 without any trend, normalised per stream with the factor recorded, and the two streams are weighted by their 2025 tonnages:

$$S_j = \frac{M_{2025}^d S_j^{(d)} + M_{2025}^n S_j^{(n)}}{M_{2025}^d + M_{2025}^n}, \quad S_j^{(k)} = x_j^{(k)} / \sum_i x_i^{(k)} \quad (2)$$

The 12 RIPS categories (plus glass) are mapped to ten IPCC-aligned model fractions. Two harmonisation rules handle inconsistent category definitions: where organics are reported as one aggregate, a share γ (median 0.27, range 0.04–0.62 from the locations that report both) is moved to garden waste, and where wood is at least 10% and no leaf category exists, it is treated as garden waste. Managed waste, which determines plant size in the managed-tonnage scenarios, is held in two series that are never merged silently: the RIPS status index (M1, 19 locations) and a series that uses local service or handling indicators where available (M2). The two differ by up to 86 percentage points for the same location (Fig. 1). A derived share for Padang (93.7%) depends on a recovery figure that may be a potential, so only its disposal-only lower bound (71.8%) enters M2; a 14% target for Serang is treated as a scenario, not a baseline.

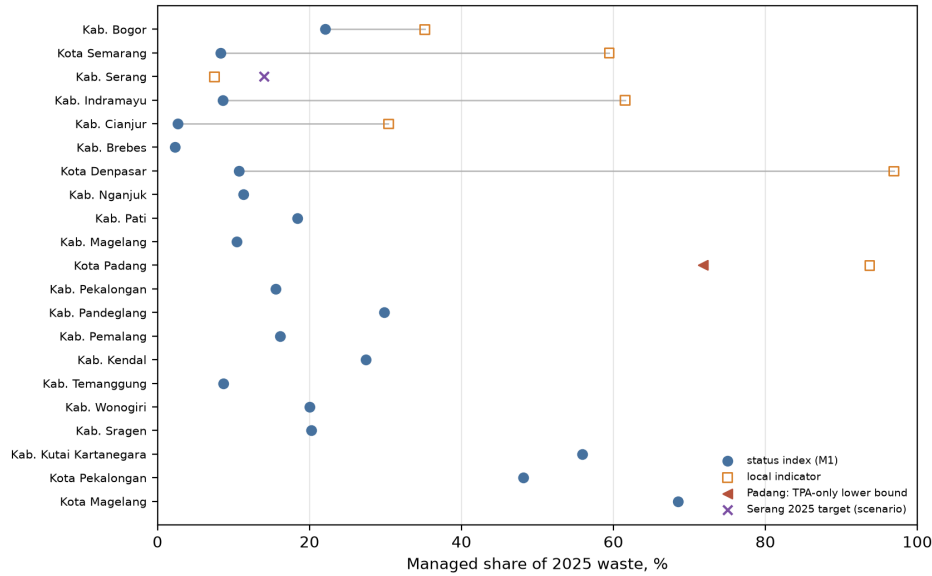


Fig. 1. Managed share of 2025 waste by indicator family. The RIPS status index and local indicators measure different things; grey lines join values for the same location.

2.3 Waste-fraction taxonomy and material parameters

Following the tiered approach of Edjabou et al. (2015), the database holds 10 Level I groups, 36 Level II fractions and a provisional catalogue of 56 Level III refinements, keyed by (level, code). Level II/III compositions obtained by allocating RIPS aggregates with priors are retained for future calibration but are not used in the computation, because several priors have no Indonesian or literature basis. Danish benchmark compositions were converted to conditional shares only for complete sibling sets on the same basis (11 sets). Parameters of the ten model fractions are the IPCC (2006) Table 2.4 defaults for DOC, total carbon and fossil share and the IPCC (2019) Table 3.0 DOCf values (0.70 food and garden, 0.50 paper and textile, 0.10 wood). DOC is on a dry basis and multiplies dry mass, so moisture is applied once ($\text{DOC}_{\text{wet}} = \text{DOC}_{\text{dry}} (1 - w)$). Rubber and leather have no verified DOCf; the gap is sampled between 0 and the generic IPCC default of 0.5. Plastic uses the aggregate median LHV of 30.5 MJ/kg dry solids reported by Götze et al. (2016); other dry-matter heating values and the RDF transfer coefficients are legacy assumptions. Moisture as received (food 0.75, plastic 0.22, paper 0.35) is an assumption calibrated to bulk moisture measured at Indonesian transfer points (Prabowo et al., 2019); IPCC moisture, which describes waste as generated, is run as a scenario. At Level II/III a DOC value is known for only 83–96% and a DOCf for 52–85% of the wet mass, which is why the computation remains at the ten model fractions (Fig. 2).

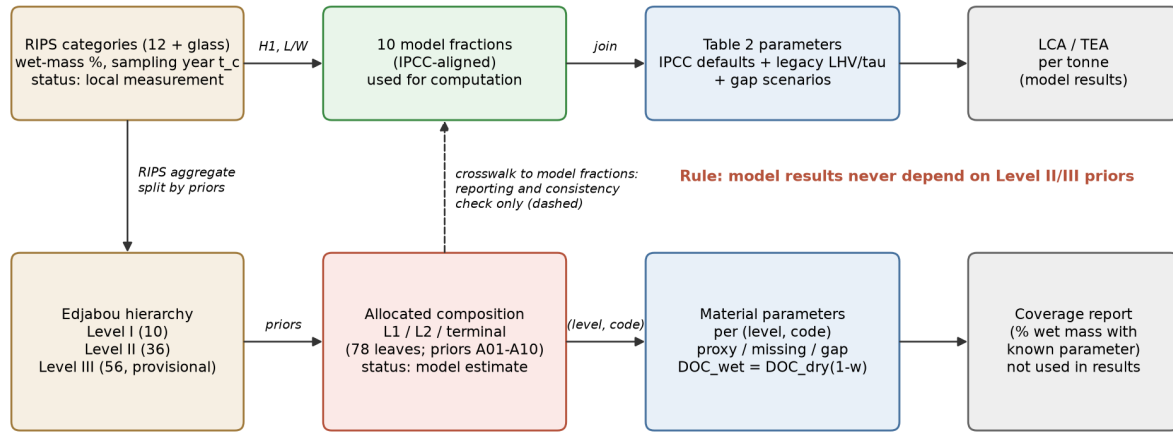


Fig. 2. Metamodel of the data and computation. The upper chain produces results; the Level I/II/III chain is reporting and calibration content that cannot alter results.

2.4 Life-cycle inventory and cost model

For a stream with fraction masses m_j per tonne MSW, methane generated in a landfill, fossil CO_2 on combustion and the lower heating value as received are

$$\text{CH}_4 = 10^3 k_D \text{MCF} F_{12}^{16} \sum_j m_j (1 - w_j) \text{DOC}_j \text{DOC}_f, \quad E_{\text{fos}} = 10^{344} \sum_j m_j (1 - w_j) C_j \varphi_j, \quad H = \sum_j k_{h,j} S_j h_j (1 - w_j) - \lambda \sum_j S_j w_j \quad (3)$$

The landfill module applies $\text{MCF} = 0.8$ and no gas collection for the open dump, and $\text{MCF} = 1$, lifetime collection efficiency $\eta = 0.5$ (0.3–0.8) and 10% cover oxidation for the sanitary landfill. WtE uses a net electrical efficiency of 0.18 (0.14–0.22). The RDF line sorts a fixed share of each fraction, dries the product to 20% moisture with its own fuel, hauls it to the nearest kiln and credits displaced coal at 0.9 GJ per GJ; rejects are landfilled. AD captures 70% of food waste, converts it with a realised yield of $0.36 \text{ Nm}^3 \text{ CH}_4/\text{kg VS}$ and 5% fugitive loss, and charges a front-end separation step as 40% of the RDF-line cost. PHB is produced from captured gas at $2.3 \text{ t CH}_4/\text{t PHB}$ and credited against polypropylene. Costs combine annualised CAPEX scaled with capacity to the power $b = 0.7$, O&M, residue landfill cost that falls with landfill size, haulage and product revenue. All 61 parameters, their ranges and their sources are listed in the assumption register (Supplementary Table S3).

Two further indicators reuse the same inventory. Fossil cumulative energy demand (CED) converts grid electricity with a primary-energy factor of $3.6/\eta \text{ MJ/kWh}$ ($\eta = 0.32$, range 0.28–0.36, plus fuel supply), diesel-type ancillary burdens with the IPCC diesel factor, displaced kiln coal by its energy content plus supply, and displaced polypropylene with its cradle-to-gate CED; biogenic energy is not counted (Frischknecht et al., 2015). Land take is the landfill area consumed per tonne, $1/(\rho H)$ times a gross-area factor ($\rho = 0.8 \text{ t/m}^3$, $H = 20 \text{ m}$ for a sanitary landfill; 0.5 t/m^3 and 5 m for an open dump), plus plant footprints over their lifetime. Most land-use factors are engineering assumptions and are sampled over wide ranges.

2.5 Decision analysis, uncertainty and scenarios

An option is removed where it fails a gate: WtE requires $\text{LHV} \geq 7 \text{ MJ/kg}$ and at least 150 t/day (Rand et al., 2000); RDF requires a net calorific value of at least 12.56 MJ/kg and a kiln within 300 km by road; PHB requires at least 500 t/yr. Among feasible options, the social cost $\text{SC}_k = C_k + p G_k/10^3$ is computed for a carbon value p sampled uniformly between 0 and 100 USD/t CO_2e , following the logic of stochastic multicriteria acceptability analysis (Lahdelma and Salminen, 2001). The probability of being best is the share of 4,000 Monte Carlo draws per location in which an option has the lowest social cost. Composition is sampled from a Dirichlet distribution $S \sim \text{Dir}(\alpha_0 s)$ with heuristic concentrations ($\alpha_0 = 80$, or 40 for flagged data) that are not statistically calibrated; parameters are sampled from triangular distributions. The two sources were also sampled separately to show their contributions. Ten scenarios were run with common random numbers: the market case, the Perpres 109/2025 tariff, food separated at source, residues to an open dump, the two managed-tonnage series, GWP20, IPCC moisture, projected overview tonnage and rubber DOCf of 0.5. Sensitivity was measured by Spearman rank correlation within the Monte Carlo sample and by Sobol indices (Saltelli et al., 2010) for every location.

2.6 Scenario discovery

To answer the research question in terms of conditions rather than cases, the model was run over 40,000 combinations of conditions a planner can observe or choose: carbon value (0–100 USD/t CO₂e), waste to the facility (50–3,000 t/day), road distance to a cement kiln (10–400 km), grid region, availability of the WtE tariff, source separation of food, composition, moisture, landfill-gas collection and landfill cost, with the other parameters sampled from the register. Composition was drawn from a Dirichlet distribution centred on the mean RIPS composition with a concentration of 46 fitted to the between-city spread. The best feasible option of each draw was explained with a classification tree of depth 4 (Breiman et al., 1984) and with PRIM boxes (Friedman and Fisher, 1999), following the scenario-discovery approach of decision making under deep uncertainty.

2.7 Verification

An automated suite of 60 checks (all passed) tests blank-versus-zero handling, uniqueness of taxonomy keys, domestic plus non-domestic mass balance, parent-child sums at every level, the single projection to 2025, managed tonnage not exceeding total tonnage, wet/dry consistency (the dry DOC and IPCC moisture reproduce the IPCC wet DOC within 0.005) and the presence of a source, locator and status for every value. Model outputs were compared with published Indonesian measurements (Table 1).

3. Results

3.1 Waste characteristics and feasibility

Quantity	Model (21 locations)	Reference
bulk moisture (-)	0.39 to 0.60 (median 0.51)	0.53 to 0.56 at transfer points (Prabowo et al. 2019); 0.64 to 0.66 at a landfill (Fiki et al. 2022)
LHV as received (MJ/kg)	5.0 to 9.5 (median 7.2)	about 5 to 7 (HHV 6.9 to 9.0 measured, Prabowo et al. 2019; 6.5 assumed, Azis et al. 2021)
HHV of dry matter (MJ/kg)	15.3 to 20.3 (median 18.1)	15.7 to 19.2, from HHV 6.9 to 9.0 as received at 53 to 56% moisture (Prabowo et al. 2019)
L0 at MCF = 1 (kg CH ₄ /t)	34 to 65 (median 45)	no Indonesian measurement; reported for transparency
RDF yield (t/t MSW)	0.17 to 0.38 (median 0.27)	0.20 to 0.42 (Indonesian RDF plants, GIZ 2023)
RDF heating value (MJ/kg)	15.0 to 19.8 (median 17.5)	12.6 to 13.8 for biodried whole waste (GIZ 2023); sorted and dried SRF is higher
WtE net electricity (kWh/t)	248 to 474 (median 360)	632 kWh/t at 35% gross conversion (Azis et al. 2021); model uses 18% net
carbon in biogas / carbon in food	0.74	must be below 1

Table 1. Characteristics of the harmonised 2025 waste compared with published Indonesian values.

Bulk moisture as received ranged from 0.39 to 0.60 and LHV from 5.0 to 9.5 MJ/kg (median 7.2), at the dry and energetic end of measured Indonesian values (Table 1). 12 of 21 locations passed the WtE heating-value gate and 11 passed both the heating-value and the supply gate. 6 locations generate at least 1,000 t/day, but on managed waste none does in series M1 and only Kab. Bogor does in series M2. Methane potential at MCF = 1 was 34–65 kg CH₄/t, and the RDF yield 0.17–0.38 t/t, within the range reported for Indonesian RDF plants.

3.2 Climate and cost performance

Option	G	ΔG vs open dump	ΔG vs landfill	C	MAC vs OD	MAC vs SL	Pass
SL landfill + flare	557 (424 to 793)	425 (322 to 609)	-	18 (12 to 40)	34 (21 to 81)	-	21
S1 WtE	121 (-27 to 241)	878 (546 to 1404)	439 (206 to 795)	58 (42 to 120)	62 (36 to 131)	90 (47 to 183)	11
S2 RDF	319 (136 to 421)	661 (455 to 998)	251 (110 to 390)	33 (23 to 61)	43 (31 to 75)	64 (28 to 113)	20
S3 AD	373 (174 to 688)	594 (460 to 775)	173 (104 to 250)	31 (21 to 61)	46 (30 to 97)	69 (40 to 149)	21
S4 PHB	543 (414 to 773)	439 (332 to 628)	14 (10 to 20)	38 (23 to 75)	78 (54 to 154)	1449 (993 to 2412)	20
S5 RDF + AD	142 (13 to 296)	826 (654 to 1106)	420 (309 to 525)	36 (25 to 67)	37 (29 to 70)	42 (25 to 62)	20

Table 2. Results per tonne of MSW at central values, market case: median (range) over 21 locations. G and ΔG in kg CO₂e/t, C in USD/t, MAC in USD/t CO₂e. Pass: number of locations passing the gates.

Against an open dump, WtE avoided 878 (546–1404) kg CO₂e/t and RDF + AD 826 (654–1106); RDF alone 661 (455–998) and AD alone 594 (460–775), because each leaves part of the waste for the landfill. A sanitary landfill with flare already avoided 425 (322–609). Net costs were 18 (12–40) USD/t for the landfill, 31 (21–61) for AD, 33 (23–61) for RDF, 36 (25–67) for RDF + AD and 58 (42–120) for WtE at market electricity value. PHB avoided only 14 (10–20) kg CO₂e/t relative to the landfill it is built on, at an abatement cost of 1449 (993–2412) USD/t CO₂e (Table 2; Fig. 3).

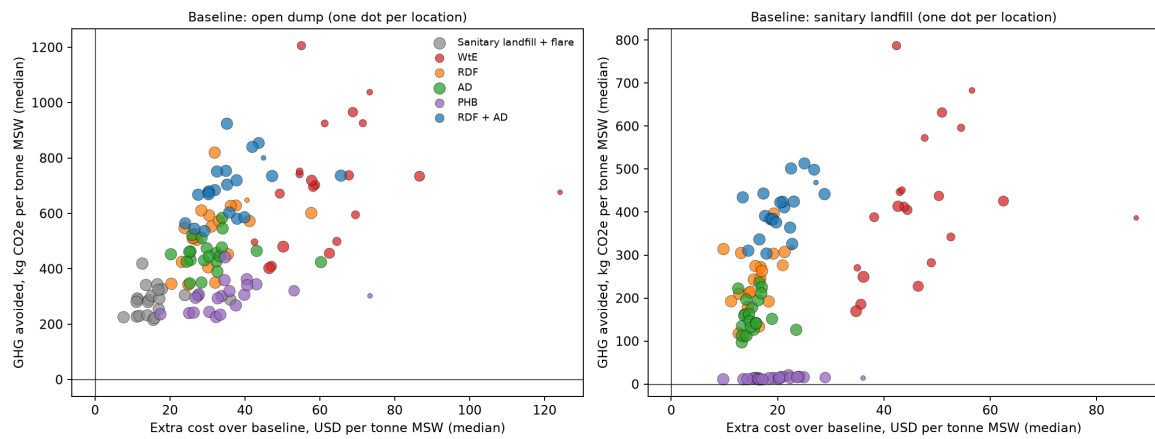


Fig. 3. GHG avoided against extra cost relative to the open dump (left) and the sanitary landfill (right); Monte Carlo medians, one point per location and option; point size increases with the probability of passing the gates.

3.3 Preferred options

At central values the sanitary landfill had the lowest social cost in all 21 locations at carbon values of 0 and 25 USD/t CO₂e. RDF + AD was best in 18 locations at 50 USD/t and 20 at 100 USD/t, while WtE was best only in Kab. Kutai Kartanegara at 100 USD/t, where no kiln lies within reach for RDF. With the carbon value sampled, the landfill was the most probable choice in 17 locations and RDF + AD in 4, but in 6 locations the lead was below 0.05 and both options are statistically close (Fig. 4). Averaged over locations, RDF + AD was on the Pareto front in 94% of draws and the landfill in 95%; RDF + AD was best in 67% of draws with carbon values of 50–100 USD/t.

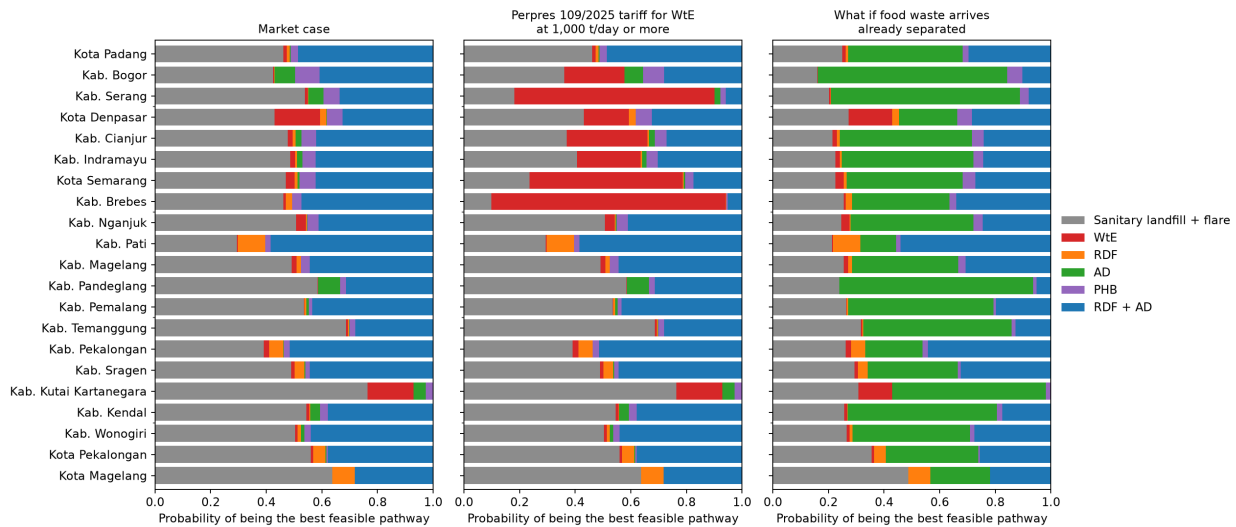


Fig. 4. Probability of being the best feasible option in the market case, with the Perpres 109/2025 WtE tariff, and with food waste separated at source.

With the Perpres 109/2025 tariff, WtE became the most probable option in 3 locations (Kab. Serang, Kota Semarang and Kab. Brebes). Brebes qualifies on generated waste only, of which 2.3% is managed according to the status index, and Semarang's tonnage is an overview total of unverified year. If food waste arrived separated, AD alone became most probable in 16 locations; for mixed waste the cost of separating food removes its advantage. On managed tonnage the landfill was the most probable option in 18 of 19 locations (M1) and 20 of 21 (M2): at the tonnages managed today every recovery plant is small and expensive.

3.4 Robustness to data provenance

Projecting the eight overview totals from their composition year and raising rubber DOCf to 0.5 changed no most-probable option (Fig. 5). The moisture assumption did: with IPCC default moisture, RDF + AD became the most probable option in 20 locations, because drier waste carries more degradable carbon into the landfill and more energy into the RDF. A 20-year horizon for methane had a similar effect (20 locations for RDF + AD and 1 for WtE). Sending residues to an open dump, which is illegal, made AD and RDF + AD cheaper in several locations because dumping residues is cheap.

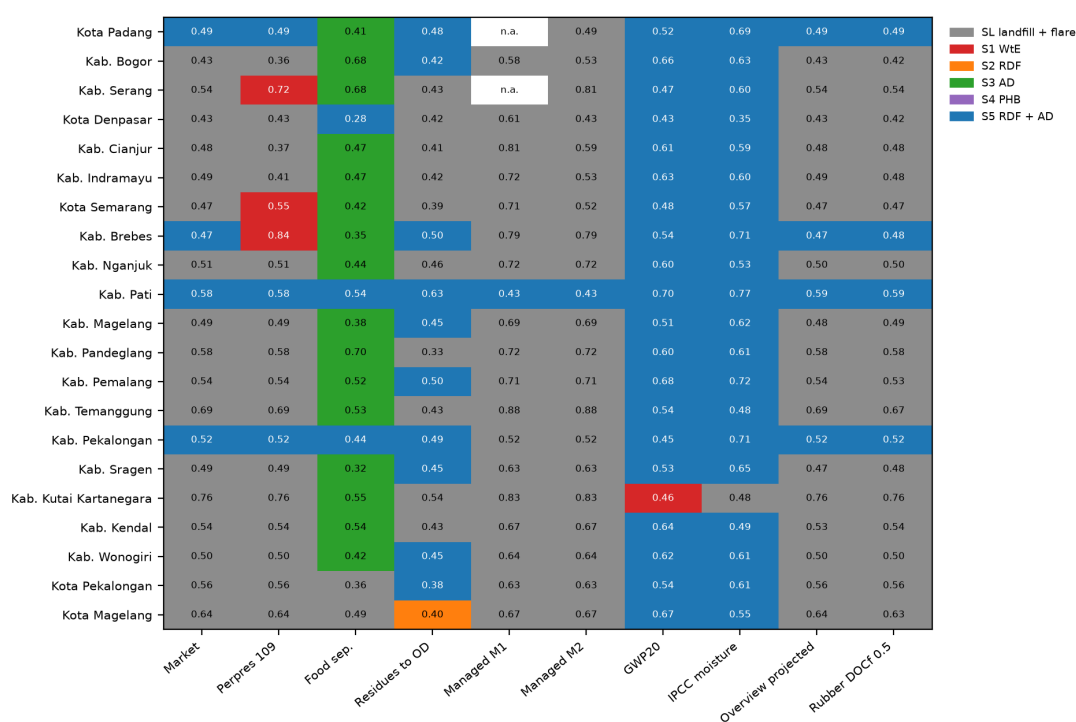


Fig. 5. Most probable option per location in each scenario (colour) and its probability of being best (number). n.a.: no status-index value for the managed-tonnage series M1.

3.5 Drivers of uncertainty

For climate results, the lifetime landfill-gas collection efficiency dominated every option that landfills residues, followed by the plastic share (fossil CO₂ in WtE), the garden share and moisture; for cost, the landfill cost, WtE and AD capital costs, RDF operating cost, the AD front-end share and PHB cost and price dominated (Supplementary Fig. S1). Sampling composition alone produced a median 90% interval of 210 kg CO₂e/t for WtE against 138 for parameters alone, whereas for the landfill parameters dominated (494 against 203). Composition hardly affected cost.

Sobol indices give the same picture: moisture as received (total index 0.28) and landfill-gas collection (0.23) explain about half of the variance of the social-cost gap between RDF + AD and the landfill at 50 USD/t CO₂e, followed by RDF operating cost and price.

3.6 Fossil energy and land take

Option	CED, GJ/t: median (range)	Saving vs open dump	Land take, m ² /t: median (range)	Saved vs open dump
SL landfill + flare	0.08 (0.08 to 0.08)	-0.06	0.081 (0.081 to 0.081)	0.319
S1 WtE	-4.20 (-5.58 to -2.84)	4.21	0.019 (0.019 to 0.019)	0.381
S2 RDF	-3.80 (-5.77 to -1.81)	3.82	0.061 (0.053 to 0.066)	0.339
S3 AD	-0.47 (-0.77 to -0.19)	0.48	0.067 (0.059 to 0.073)	0.333
S4 PHB	-0.57 (-0.85 to -0.42)	0.59	0.081 (0.081 to 0.081)	0.319
S5 RDF + AD	-4.49 (-6.29 to -2.78)	4.50	0.045 (0.042 to 0.051)	0.355

Table 3. Fossil cumulative energy demand and land take per tonne of MSW, central values, market case.

RDF + AD, WtE and RDF saved 4.5, 4.2 and 3.8 GJ of fossil energy per tonne by displacing grid electricity and kiln coal, AD alone 0.5 GJ and PHB 0.6 GJ (Table 3). An open dump took about 0.40 m² of land per tonne and a sanitary landfill 0.081 m²; RDF + AD reduced this to 0.045 and WtE to 0.019 m²/t. Energy and climate rankings agree; land take is an additional argument for diversion where land is scarce.

3.7 Conditions under which each option becomes preferable

Rule (conditions)	Best option	Purity	Share
food separated = yes and carbon value > 53.2 and Q tpd <= 1e+03	S3 AD	0.69	17%
food separated = no and carbon value <= 39.6 and Q tpd <= 1e+03	SL landfill + flare	0.95	15%
food separated = no and carbon value > 55.6 and kiln km <= 299 and Q tpd <= 1.02e+03	S5 RDF + AD	0.64	12%
food separated = yes and 29.2 < carbon value <= 53.2 and Q tpd <= 1e+03	S3 AD	0.63	8%
food separated = no and 39.6 < carbon value <= 55.6 and Q tpd <= 1e+03	SL landfill + flare	0.75	6%

Rule (conditions)	Best option	Purity	Share
food separated = yes and $13.5 < \text{carbon value} \leq 29.2$ and $Q_{\text{tpd}} \leq 979$	SL landfill + flare	0.63	6%
food separated = yes and carbon value > 29.2 and $Q_{\text{tpd}} > 1\text{e}+03$ and $\text{LHV} > 7$	S1 WtE	0.52	5%
food separated = yes and carbon value ≤ 13.5 and $Q_{\text{tpd}} \leq 933$	SL landfill + flare	0.91	5%
food separated = no and carbon value > 55.6 and kiln km ≤ 299 and $Q_{\text{tpd}} > 1.02\text{e}+03$	S5 RDF + AD	0.52	4%
food separated = yes and carbon value > 29.2 and $Q_{\text{tpd}} > 1\text{e}+03$ and $\text{LHV} \leq 7$	S3 AD	0.75	4%

Table 4. Leaves of the decision tree covering at least 4% of the 40,000 draws. The tree reproduces the best option in 70% of held-out draws (majority-class baseline 43%). Purity: share of draws in the leaf where the predicted option is best.

Carbon value, source separation of food, scale and kiln distance carried the decision (tree importances 0.41, 0.31, 0.12 and 0.08). For mixed waste without the WtE tariff the landfill was best in 62% of draws and RDF + AD in 27%. The landfill was preferred below about 40 USD/t CO₂e (purity 0.95) and, at any carbon value, where no kiln lay within about 300 km. RDF + AD was preferred above about 55 USD/t with a kiln in reach; PRIM located its box at carbon values above 51 USD/t, kiln distances below 296 km and gas collection below 0.68. WtE won almost only in the box defined by the tariff, at least 1,000 t/day and an LHV of at least 7 MJ/kg (density 0.97). AD alone won where food arrived separated and carbon was valued above about 30 USD/t. RDF alone and PHB were best in fewer than 3% of draws (Table 4; Fig. 6).

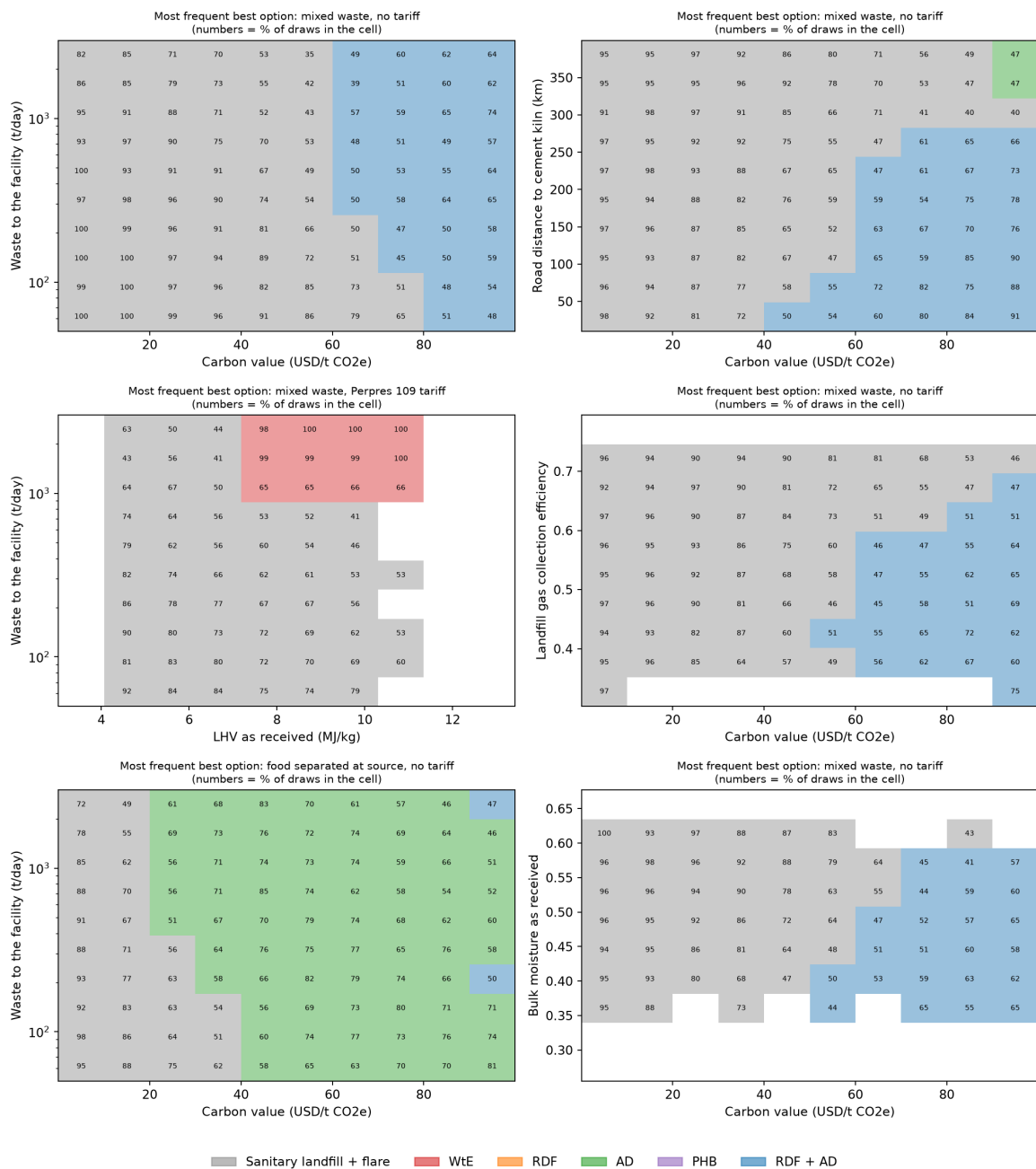


Fig. 6. Condition maps: most frequent best option and its frequency (%) on grids of two conditions, for mixed waste without tariff, mixed waste with the Perpres 109/2025 tariff, and food separated at source.

4. Discussion

4.1 Conditions under which each pathway becomes preferable

The scenario discovery of Section 3.7 states the answer to the research question compactly; the 21 locations illustrate it. The screening identifies conditions rather than a single winner. A well-operated sanitary landfill with flaring is the least-cost compliant option wherever carbon is valued below about 25 USD/t CO₂e, which includes Indonesia's current carbon tax of about 2 USD/t (UU 7/2021). Integrated RDF + AD becomes preferable once carbon is valued at roughly 42 USD/t, provided a cement kiln is within reach and the plant receives enough waste; it is the fair counterpart of WtE because both treat the whole tonne, yet it costs less and passes the RDF quality gate in most locations. WtE requires the conjunction of a heating value above 7 MJ/kg, a supply of at least 1,000 t/day and the Perpres 109/2025 tariff; without the tariff it is preferred only where no kiln is accessible. AD alone is a pathway for source-separated food and market waste, not for mixed waste. PHB from landfill gas adds little climate benefit over flaring and does not cover its production cost at current prices, consistent with techno-economic assessments of methane-based PHB (Levett et al., 2016).

4.2 Why provenance matters

The data scenarios show which provenance decisions are consequential. The year assigned to overview tonnage totals changes plant scale by up to a fifth but no decision, and the unresolved DOCf of rubber and leather is immaterial because the fraction is small. Moisture, in contrast, is decisive. IPCC moisture values describe waste as generated; waste arriving at a facility in a humid climate is wetter, and the as-received values used here bring bulk moisture into the measured range (Prabowo et al., 2019). Because DOC, fossil carbon and heating value all multiply the same dry mass, moisture moves the landfill baseline and the recovery options in opposite directions. Managed-waste indicators are equally consequential for policy: the same location may report 11% or 97% depending on whether a status index or a reduction-plus-handling indicator is used, so the eligibility of a location for a 1,000 t/day WtE plant on managed waste depends on a definition rather than on the waste. Reporting both series, instead of filling gaps with numerically similar values, keeps this visible.

The separation between the computation and the Level II/III hierarchy is a deliberate design choice. Fine-grained taxonomies are valuable for planning recycling and for designing sorting campaigns (Edjabou et al., 2015), but in the absence of local sorting data their sub-fraction shares are priors, and their properties are parent-category proxies. Allowing such priors to drive results would add apparent resolution without information. The database therefore records where the gaps lie, for example the material composition of coated cartons, composite films, batteries and WEEE, and the share of wet mass for which a parameter is known, so that future measurements can be added without changing the model.

4.3 Implications for local planning

Three practical implications follow. First, upgrading open dumps to sanitary landfills with gas collection is the largest and cheapest single step in climate terms (about 425 kg CO₂e/t at about 18 USD/t), and the quality of gas collection is the parameter that most influences every comparison. Second, recovery investment decisions should be conditioned on an explicit carbon value and on access to a cement kiln, rather than on technology preference. Third, before committing to WtE under Perpres 109/2025, local governments should verify heating value by proximate analysis of waste as received and confirm the tonnage that is actually collected, because both gates are close to their thresholds in several locations.

4.4 Limitations

The study is a screening assessment. Compositions sampled between 2014 and 2025 represent 2025 without a trend, and eight tonnage totals have an unverified year. Fraction-level moisture and heating values have not been measured for the study locations; several heating values and the RDF transfer coefficients are legacy assumptions, and some literature values remain to be checked against their originals. Only climate change is assessed; landfill gas is assigned without time dynamics; costs are class-5 estimates without escalation; coordinates are provisional; inputs are treated as independent; and 12 of the 21 locations are in Central Java. The Dirichlet concentrations are heuristics, and the Danish benchmarks describe residual household waste, not Indonesian mixed MSW. Land-use factors and several CED factors are assumptions, and the decision rules approximate the model (70% accuracy) within the range of observed compositions. The next steps are local proximate analysis, a common managed-waste definition across RIPS documents, and full inventories in a dedicated LCA software to extend the assessment to other impact categories.

5. Conclusions

A provenance-aware screening of six MSW options for 21 Indonesian cities and regencies shows that, with equal system boundaries, a sanitary landfill with flaring is the least-cost option at low carbon values, integrated RDF + AD becomes preferable at carbon values of about 40–50 USD/t CO₂e where a cement kiln is accessible, WtE depends on the Perpres 109/2025 tariff and large generated tonnage, AD alone suits source-separated food waste, and PHB from landfill gas is not competitive. Scenario discovery condenses these findings into rules that planners can check against their own carbon value, scale, kiln access, tariff and source separation. Labelling every input by its evidence type, projecting tonnage once and keeping managed-waste definitions apart did not overturn these conditions, but it revealed that the moisture of waste as received can. Measured as-received moisture and heating value, and a harmonised definition of managed waste, are the data investments that would most improve waste-infrastructure decisions in Indonesian cities.

Data and code availability

The database, model, notebook, validation suite and the scripts that generate this manuscript are provided in the MSW_pathway_model_v3 package (build_database_v3.py, msw_pathway_model_v3.py, MSW_pathway_model_v3.ipynb, validate_v3.py). Row-level RIPS source locators are given in input_kota_2025_updated.csv. The package mswpath and the notebook MSW_Decision_Tool_Colab.ipynb let other researchers and planners run the model in Google Colab for their own cities from a one-row-per-city template, re-run the scenario discovery and download Excel/HTML reports in English or Indonesian.

CRedit authorship contribution statement (draft)

Muhammad Fachri Ridwan: Conceptualization, Data curation, Methodology, Software, Formal analysis, Writing – original draft. Anthony Halog: Conceptualization, Supervision, Writing – review and editing. To be confirmed by the authors.

Declaration of generative AI in the writing process

During preparation of this draft the authors used Claude (Anthropic) to restructure the model code, generate the data pipeline and draft text from model outputs. The authors must review and edit the content and take full responsibility for the publication.

Declaration of competing interest

To be completed by the authors.

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Supplementary material

Table S1. Harmonised 2025 inputs and gates by location (central values).

Location	t_c	Flags	Q2025 t/d	M	LHV	L0 SL	RDF NCV	km	G1	G3	Most probable (p)
Kota Padang	2023	N	656	0.52	7.2	46	18.1	17	yes	no	S5 (0.49)
Kab. Bogor	2022	G	2,891	0.54	5.9	36	17.9	20	no	yes	SL (0.43)
Kab. Serang	2024	G	1,187	0.52	7.9	37	19.8	118	yes	yes	SL (0.54)
Kota Denpasar	2023	G	918	0.51	6.7	65	15.5	253	no	no	SL (0.43)
Kab. Cianjur	2023	LG	1,114	0.53	6.0	45	16.7	51	no	yes	SL (0.48)
Kab. Indramayu	2023	LG	1,152	0.52	5.7	45	16.4	55	no	yes	SL (0.49)
Kota Semarang	2022	LT	1,190	0.48	7.1	43	17.5	152	yes	yes	SL (0.47)
Kab. Brebes	2025	T	1,033	0.41	8.6	36	19.2	80	yes	yes	S5 (0.47)
Kab. Nganjuk	2022	T	744	0.54	6.0	54	16.7	106	no	no	SL (0.51)
Kab. Pati	2024	R	691	0.39	9.5	37	17.8	64	yes	no	S5 (0.58)
Kab. Magelang	2016	T	671	0.48	7.3	44	16.9	167	yes	no	SL (0.49)
Kab. Pandeang	2023		527	0.54	6.7	34	18.2	94	no	no	SL (0.58)
Kab. Pemalang	2025		484	0.50	7.9	36	18.4	89	yes	no	SL (0.54)
Kab. Temanggung	2022		423	0.45	6.6	40	17.7	158	no	no	SL (0.69)
Kab. Pekalongan	2025	L	560	0.49	8.3	48	17.7	93	yes	no	S5 (0.52)
Kab. Sragen	2016	LT	402	0.49	7.4	46	17.3	105	yes	no	SL (0.49)
Kab. Kutai Kartanegara	2021	WT	373	0.50	7.2	56	16.6	331	yes	no	SL (0.76)
Kab. Kendal	2014	WT	437	0.60	5.0	57	15.0	177	no	no	SL (0.54)
Kab. Wonogiri	2022	WRT	422	0.56	6.0	53	15.7	158	no	no	SL (0.50)
Kota Pekalongan	2024	L	164	0.51	7.8	48	17.9	115	yes	no	SL (0.56)
Kota Magelang	2024	W	50	0.46	8.1	48	17.4	165	yes	no	SL (0.64)

Flags: L lumped organics, W wood as garden, R rounded percentages, G glass in residual, N glass not reported, T tonnage year unverified. M bulk moisture; LHV in MJ/kg; L0 in $\text{kg CH}_4/\text{t}$ at $\text{MCF} = 1$; km road distance to kiln; G1 WtE heating-value gate; G3 PSEL scale on generated waste.

Table S2. Managed waste by indicator family: see Table 5 of the methodology (MSW_Methodology_v3_EN.pdf).

Table S3. Assumption register: data/assumption_register.csv and Table 3 of the methodology.

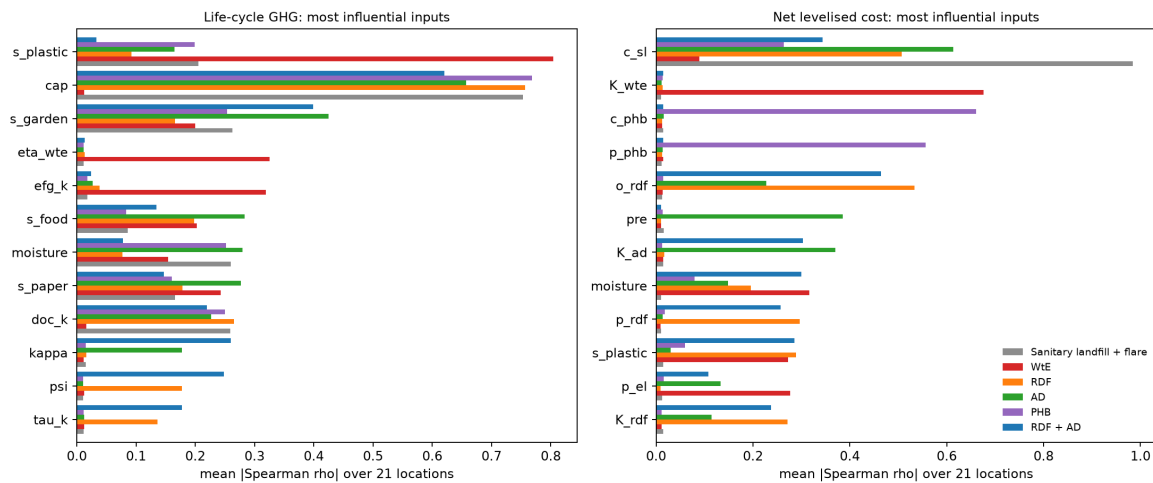


Fig. S1. Mean absolute Spearman rank correlation between inputs and results over the 21 locations.